A) For M1

```
Create matrix M[n][n]
for i ← 0 to n-1 do
    for j ← 0 to n-1 do
        M[i][j] ← 5 + n*i + j
```

↳ from equation derived from the matrix

$$M[x, y] = 5nx + y$$

B) For M_2

```
Create matrix M[n][n]
 value ← 5
for diagonal d ← 0 to 2n-2 do
    if d is even then
        row ← min(d, n-1)
        col ← d - row
    while row >= AND col < n do
        M[row][col] ← value
        value ← value+1
        row ← row-1
        col ← col+1
    Else
        col ← min(d, n-1)
        row ← d - col
    while col >= 0 AND row < n do
        M[row][col] ← value
        value ← value+1
        row ← row+1
        col ← col-1
```

3) For M:3

Create matrix M[n][n]
for i ← 0 to n-1 do
  for j ← 0 to n-1 do
    M[i][j] ← 5+i+n*j
from the equation
M(x,y) = 5+x+ny

B) Algorithm SearchSS(M, key)
i ← 0
j ← n-1
while i<n AND j>=0 do
  if M[i][j] = key then
    print(i,j)
    break
  Else if key < M[i][j] then
    j ← j-1
  Else
    i ← i+1
  print "Not Found"

B1) Time Complexity
- One Comparison per step which each step eliminates one row or one column
worst case 2n-1 comparisons
→ O(n)

B2) Space Complexity

C1) Algorithm DACSearch2S(M, key, top, bottom, left, right)

```
if top > bottom OR left > right then
    return not found
midRow ← (top + bottom)/2
midCol ← (left + right)/2
if M[midRow][midCol] = key then
    Print (midRow, midCol)
    return
else if key < M[midRow][midCol] then
    DACSearch2S(M, key, top, midRow-1, left, midCol-1)
    DACSearch2S(M, key, top, midRow-1, midCol, right)
    DACSearch2S(M, key, midRow, bottom, left, midCol-1)
else
    DACSearch2S(M, key, top, midRow, midCol+1, right)
    DACSearch2S(M, key, midRow+1, bottom, left, midCol)
    DACSearch2S(M, key, midRow+1, bottom, midCol+1, right)
```

C1) Time-Complexity

Worst case recurrence check the three quadrants

$T(n) = 3T(n/2) + 1$

Applying master theorem $a = 3, b = 2$ $f(n) = 1$

result = $O(n^{\log_2 3})$

C2) Space-Complexity

D1) Mathematical Comparison

Search2S → $O(n)$ Whereas DACSearch2S > $O(n)$

Therefore Search2S is faster than DACSearch2S